

Stability of Adjoint Sampling on a One-Dimensional Gaussian Target

1 Setup

We consider the one-dimensional Gaussian problem

$$p_\star = N(1, 0.02), \quad p_0 = N(1.05, 0.02), \quad \sigma(t) = 1, \quad t \in [0, 1].$$

An *outer iteration* generates a new terminal distribution using the current drift and then updates the drift from samples of that distribution. For the learned samplers, the drift is represented by a neural network. The first network fit uses 5000 optimization steps; subsequent outer iterations are warm-started and use 100 optimization steps.

The practical Adjoint sampler retains previous endpoint samples in a *replay buffer*. The no-replay ablation uses only endpoints from the current outer iteration. Parameter damping replaces the fully trained proposal parameters $\hat{\theta}_{k+1}$ by

$$\theta_{k+1} = (1 - \eta)\theta_k + \eta\hat{\theta}_{k+1}.$$

Because p_0 is externally prescribed rather than generated by the initial network, the first learned update is left undamped.

For the matched-variance Gaussian model, the exact outer mean error obeys

$$m_{k+1} - 1 = A(m_k - 1), \quad A = 1 + \frac{\log(0.02)}{1 - 0.02} = -2.9918602.$$

Thus the exact update is strongly unstable. With scalar damping,

$$A_\eta = 1 - \eta + \eta A.$$

The value $\eta = 0.5$ gives

$$A_{0.5} = -0.9959301,$$

which lies very close to the stability boundary. The damping coefficient that cancels the scalar mean multiplier is

$$\eta_{\text{mean}} = \frac{1}{1 - A} = \frac{0.98}{-\log(0.02)} = 0.2505098.$$

We therefore refer to η_{mean} as *mean-optimal damping*; it is not chosen to optimize the variance dynamics.

2 Exact and learned affine updates

The affine oracle uses the exact Gaussian drift associated with the current mean and variance. Its mean follows the predicted unstable recurrence:

$$0.849, 1.437, -0.312, 4.818, -10.415, 34.334, -98.350, 291.930.$$

The variance remains close to 0.02 throughout. The run is stopped at outer iteration 8 when the absolute mean error first exceeds 100.

Replacing the exact drift by a neural network changes the outer dynamics substantially. After the 5000-step initial fit, the first learned rollout is close to the oracle: its three-seed average mean is approximately 0.834 versus the oracle value 0.849. Subsequent 100-step warm-started fits do not solve each affine regression problem to completion. The resulting iteration remains bounded for all 100 outer steps and develops persistent oscillatory behavior rather than the geometric growth of the exact map.

Over the final 20 iterations, the learned-affine runs have mean absolute mean error 0.348 and average terminal variance 0.0223. Thus finite inner optimization strongly modifies the unstable exact outer map even before the reciprocal-adjoint construction and replay are introduced.

Quantity	Setting
Target	$N(1, 0.02)$
Initial distribution	$N(1.05, 0.02)$
Diffusion coefficient	$\sigma(t) = 1$
Maximum outer iterations	100
First inner optimization	5000 updates
Later inner optimizations	100 updates
Training endpoints per iteration	512
Diagnostic terminal samples	10,000 per seed
Independent learned runs	3 seeds
Time discretization	500-step ql grid
Network	3 hidden layers of width 64
Optimizer	Adam, learning rate 10^{-4}
Replay capacity	1000 endpoints
Terminal-gradient clipping threshold	150

Table 1: Shared numerical settings. The affine oracle does not require network training or replay.

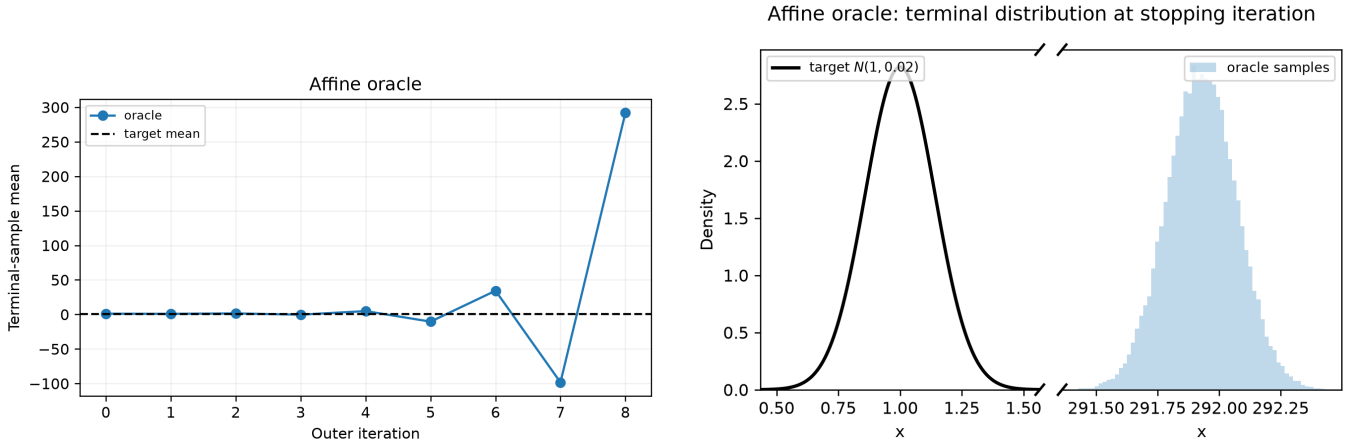


Figure 1: Affine oracle. The exact outer update exhibits rapidly growing, sign-alternating mean error while preserving the target-scale variance. The terminal histogram corresponds to the stopping iteration.

3 Full Adjoint Sampling

The full Adjoint sampler replaces direct supervision by reciprocal-adjoint matching and retains endpoints from previous outer iterations in a replay buffer. No explicit parameter damping is used in this case.

The method remains bounded through 100 outer iterations, but the three runs do not converge reproducibly to the same mean. At iteration 100 the terminal means are approximately

$$0.903, \quad 1.386, \quad 0.906.$$

Their average is 1.065, but this average hides substantial between-seed variation: the standard deviation of the three final means is 0.227. The mean absolute error over the final 20 iterations is 0.297.

The average terminal variance over the final 20 iterations is 0.0226, close to the target variance 0.02. Thus the undamped sampler reproduces the target scale more accurately than its mean location.

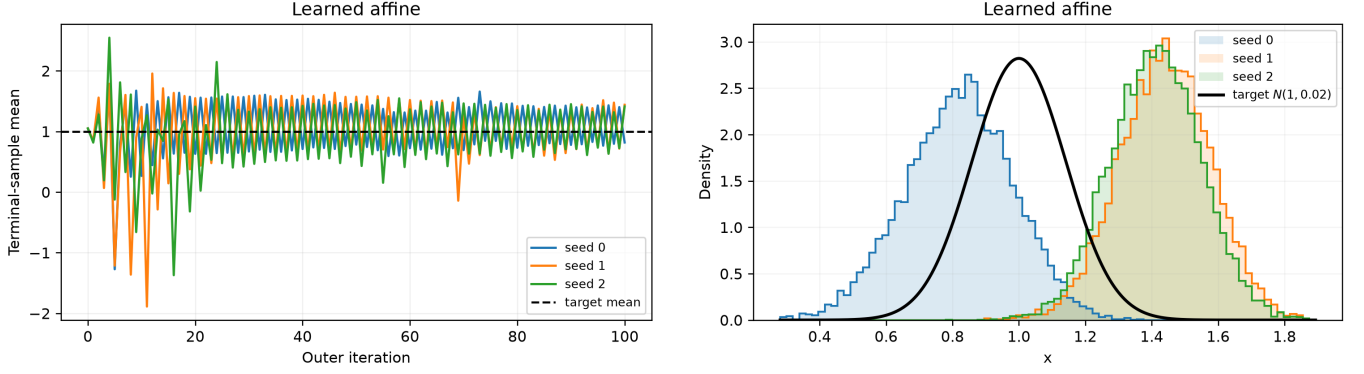


Figure 2: Learned affine update. Colors identify the same random seed in the mean trace and terminal histogram. Histograms are normalized separately rather than pooling samples from different trained networks. The black curve is the target density.

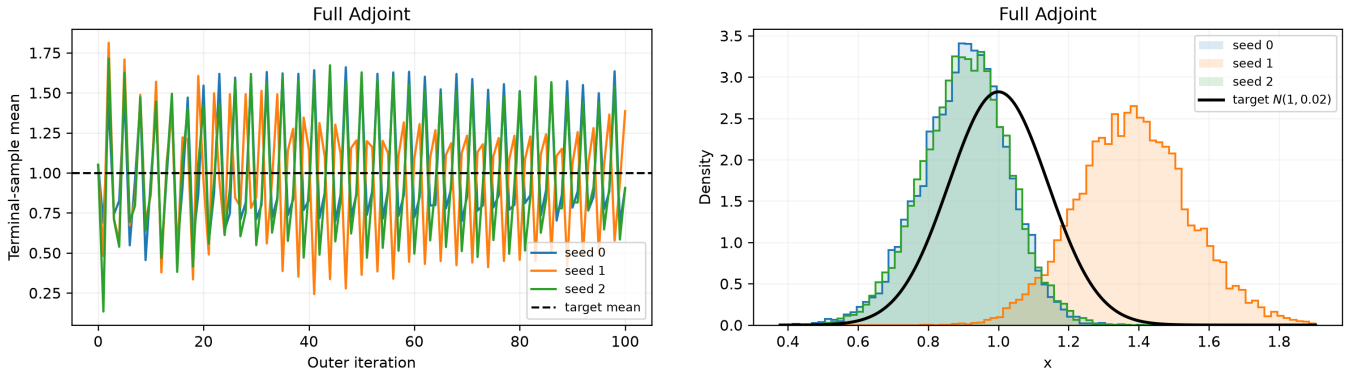


Figure 3: Full Adjoint Sampling with replay and no explicit damping. Individual seed distributions are shown separately; no cross-seed pooling is used.

4 Explicit damping

4.1 Near-boundary damping: $\eta = 0.5$

For the idealized Gaussian mean map, $\eta = 0.5$ gives $A_\eta \approx -0.996$ and therefore predicts only very weak contraction. In the learned sampler with replay, however, the three runs become much more reproducible. Their terminal means are

$$1.052, \quad 1.058, \quad 1.084,$$

with a final between-seed standard deviation of only 0.0136. The mean absolute error over the final 20 iterations is 0.0590.

This stabilization comes with a broader terminal distribution. The final average variance is 0.0367, and the final-20 average is 0.0389, both well above the target value 0.02.

4.2 Mean-optimal damping

Using

$$\eta_{\text{mean}} = 0.2505098$$

produces the strongest mean stabilization in the experiment. The three final means are

$$1.0092, \quad 0.9874, \quad 1.0131,$$

with average 1.0033 and between-seed standard deviation 0.0113. Across the final 20 iterations and all three seeds, the mean absolute error is only 0.0105.

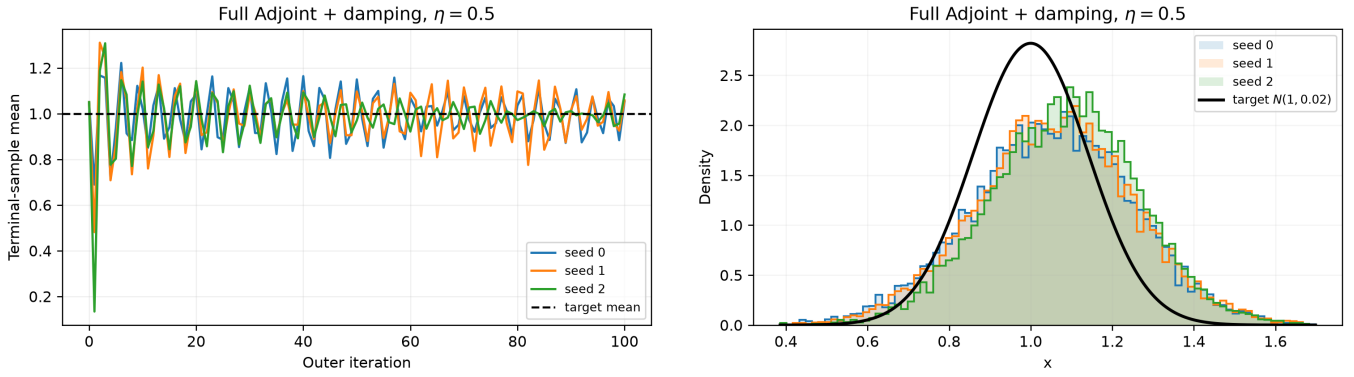


Figure 4: Full Adjoint Sampling with replay and $\eta = 0.5$ parameter damping. Mean behavior is substantially more reproducible than in the undamped sampler, while the terminal variance is inflated.

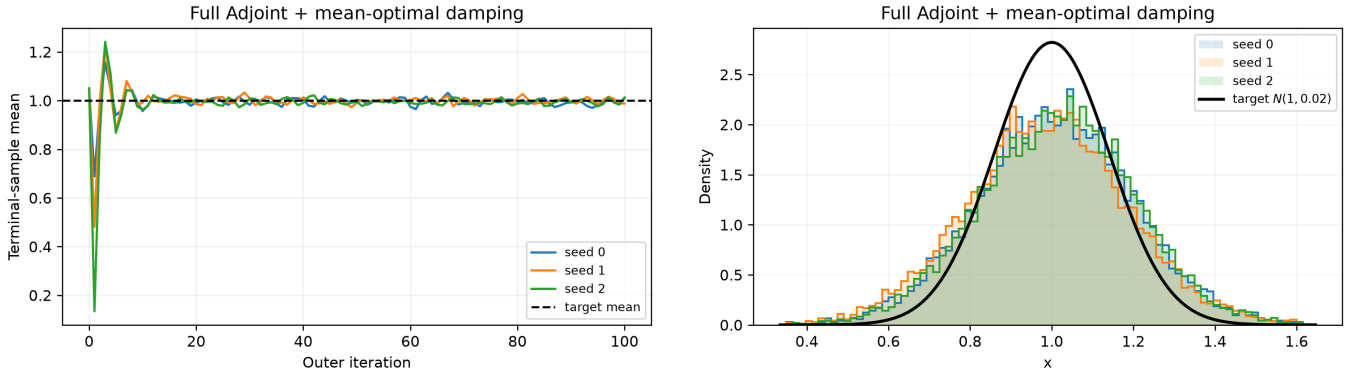


Figure 5: Full Adjoint Sampling with replay and mean-optimal damping. The terminal mean is stable across seeds and remains close to the target; the residual discrepancy is primarily in variance.

The variance again converges to a value larger than the target: the final average is 0.0370 and the final-20 average is 0.0380. Mean-optimal damping therefore removes the dominant mean instability but does not simultaneously recover the matched target variance.

5 Replay ablation

Replay has a strongly damping-dependent effect. Without explicit damping, removing replay gives a final average mean of 1.243 and a final-20 mean absolute error of 0.247. With $\eta = 0.5$, replay is decisive: the no-replay sampler has final average mean 1.811 and final-20 mean error 0.597, whereas the replay sampler has final average mean 1.065 and final-20 mean error 0.059.

For mean-optimal damping, replay has little effect on the mean. The final averages are 1.0043 without replay and 1.0033 with replay, and the final-20 mean errors are 0.0098 and 0.0105, respectively. Explicit mean-optimal damping therefore dominates the replay choice for mean stability in this example.

6 Conclusions

The Gaussian example separates three effects that are partially hidden in the full learned sampler.

First, the exact affine outer update is genuinely unstable: its mean error alternates in sign and grows at the analytically predicted rate while its variance remains near the target value. Second, replacing the exact update by a warm-started finite optimization changes the dynamical system and prevents the corresponding learned-affine iteration from exhibiting the same explosive growth; bounded oscillations remain. Third, explicit parameter damping can stabilize the mean of

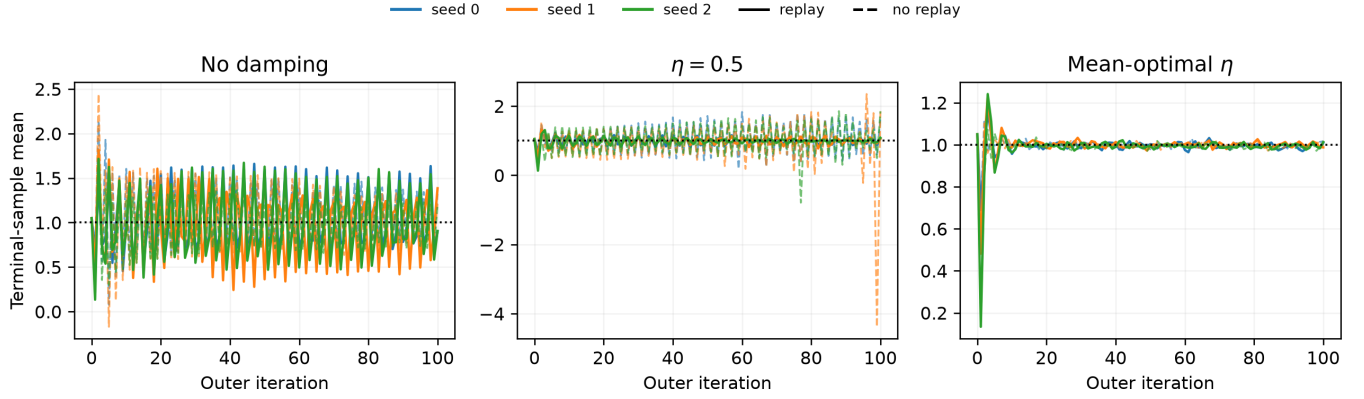


Figure 6: Replay ablation. Solid lines use replay and dashed lines use only the current outer population. Colors indicate seeds. Replay interacts strongly with $\eta = 0.5$, while its effect on the mean is small under mean-optimal damping.

Method	Final mean	Seed SD	Final variance	Final-20 $ m - 1 $
Learned affine	1.2231	0.2867	0.0215	0.3485
Full, no replay	1.2433	0.0562	0.0232	0.2470
Full + replay	1.0648	0.2268	0.0185	0.2969
Full, no replay, $\eta = 0.5$	1.8107	0.1167	0.0251	0.5975
Full + replay, $\eta = 0.5$	1.0646	0.0136	0.0367	0.0590
Full, no replay, η_{mean}	1.0043	0.0098	0.0373	0.0098
Full + replay, η_{mean}	1.0033	0.0113	0.0370	0.0105

Table 2: Summary of the learned regimes at the end of the 100-iteration run. “Seed SD” is the standard deviation of the three final terminal means. The final-20 statistic averages $|m_k - 1|$ over the final 20 outer iterations and all three seeds.

the full Adjoint sampler.

For this target, mean-optimal damping gives the clearest mean stabilization: both replay and no-replay variants remain within roughly 10^{-2} mean error over the final portion of training and show little seed dependence. The remaining error is predominantly a variance error, with terminal variance near 0.038 rather than 0.02. Thus damping derived from the scalar Gaussian mean recurrence resolves the observed mean instability but does not by itself solve the full distribution-matching problem.